

can be detected at an early age using deep learning. Genome mapping is another area being developed by deep learning since the technique can help researchers learn everything there is to know about a genome and get an early understanding of what diseases might occur in patients in the future. Another interesting application in the medical field is analysing medical insurance fraud claims with its predictive analysis.

Humanoids are also being extensively worked upon and developed which is made possible by deep learning since they need to act 'human'. The rate of growth in the sectors of robotics and deep learning is astonishing and we are not far away from a future filled with autonomous robots. A good example of this future is the robot Sophia, a humanoid created by Hanson Robotics. The great advances in Neuro-linguistic programming (NLP), as well as image processing that are enabled by deep learning, allow increasingly efficient interaction with voice assistants as well.

Other applications of deep learning are seen in fields and practices such as - space travel, drones and delivery robots, chatbots and service robots, facial recognition for surveillance, finding tumours in X-Rays, spotting genetic sequences indicating diseases, identifying molecules for more effective healthcare, predictive maintenance on infrastructure by analysing IoT sensor data, making cashier-less stores like Amazon GO possible, accurate transcription and translation and more. The list can go on and on.



Sophia, a humanoid created by Hanson Robotics, shows the potential of deep learning in robotics.

Enterprise-based applications

Deep learning is the early stages of adoption among companies across the world. Despite this, in a recent survey by O'Reilly Media hosting 3,300 technology professionals, 28 per cent were found to already be using deep learning. 54 per cent of the respondents also predicted that deep learning will play an integral role in their future projects while another 38 per cent foresee usage of deep learning in some capacity in the future.

Deep learning analysis in enterprises is often compared to older data analysis techniques. Now enterprises seem to find most utility of deep learning in making sense of structured or semi-structured data or text. Enterprises of all sizes have begun using deep learning to solve a wide range